Accrual Depreciation Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Accrual Depreciation Sensitivity (ADS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on ADS achieves an annualized gross (net) Sharpe ratio of 0.40 (0.35), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 25 (23) bps/month with a t-statistic of 2.54 (2.46), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price) is 23 bps/month with a t-statistic of 2.51.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in returns. This slow diffusion manifests particularly in accounting-based signals, where the complexity and lower salience of financial statement information create frictions in the price discovery process. Despite extensive research on accruals and their components, we lack a comprehensive understanding of how the sensitivity of accruals to depreciation policies affects future stock returns and whether investors fully incorporate this relationship into prices.

We propose that Accrual Depreciation Sensitivity (ADS) predicts stock returns through the gradual incorporation of information about firms' accounting choices and their economic implications. The slow diffusion of information framework, established by Hong et al. (2000) and Hong and Stein (1999), suggests that information spreads gradually across investors due to limited attention and processing constraints. When firms exhibit high sensitivity between their accruals and depreciation policies, this signals important information about asset quality and earnings management that requires sophisticated analysis to decode. Hirshleifer et al. (2009) demonstrate that investors exhibit limited attention to accounting information, particularly for complex signals that require combining multiple financial statement items. The relationship between accruals and depreciation represents precisely such a complex signal, as it requires investors to jointly evaluate working capital management and fixed asset accounting policies. Building on DellaVigna and Pollet (2009)'s evidence of investor inattention and Cohen and Frazzini (2008)'s findings on information processing delays, we expect that the return predictability of ADS stems from investors' gradual recognition of its implications for future profitability. This gradual information diffusion should be most pronounced among stocks with higher information processing frictions. Hong et al. (2000) show that bad news travels slowly, especially for small stocks with low analyst coverage, while Hou (2007) document that information diffuses more slowly into prices of firms with higher limits to arbitrage. Therefore, we predict that the return continuation associated with ADS will concentrate in market segments where attention constraints bind most severely—among smaller firms, those with limited analyst following, and stocks with higher idiosyncratic volatility that deters arbitrageurs.

Our empirical analysis reveals that ADS strongly predicts cross-sectional returns with economically and statistically significant results. The value-weighted long/short trading strategy based on ADS achieves an annualized gross Sharpe ratio of 0.40 and generates a monthly average abnormal gross return of 25 basis points relative to the Fama-French five-factor model plus momentum, with a t-statistic of 2.54. This performance places ADS in the top quintile of documented anomalies, with its gross Sharpe ratio exceeding 82% of strategies in the factor zoo. The strategy's robustness manifests across alternative portfolio construction methodologies, with twenty-five out of twenty-five alphas maintaining t-statistics above two across different weighting schemes and breakpoint choices. Notably, the predictive power of ADS persists even among the largest stocks, where the strategy achieves an average return of 23 basis points per month with a t-statistic of 1.76 for firms above the 80th NYSE size percentile. After accounting for transaction costs using high-frequency bid-ask spreads, the strategy maintains a net Sharpe ratio of 0.35, outperforming 93% of anomalies in the factor zoo. The net generalized alpha relative to the six-factor model equals 23 basis points per month with a t-statistic of 2.46, demonstrating that ADS expands the achievable mean-variance frontier even after implementation costs. When we control for the six most closely related anomalies—including total accruals, advertising expense, and abnormal accruals—plus the six Fama-French factors simultaneously, the ADS strategy retains an alpha of 23 basis points per month with a t-statistic of 2.51, confirming its incremental contribution to the cross-section of expected returns.

This paper makes several contributions to the asset pricing and accounting literatures by documenting a novel predictor that bridges depreciation accounting and accrual quality. While Sloan (1996) establishes the accrual anomaly and subsequent work by Richardson et al. (2005) decomposes accruals into components, our focus on the sensitivity between accruals and depreciation provides new insights into how accounting policy interactions predict returns. Unlike Xie (2001), who examine abnormal accruals through discretionary components, we capture the systematic relationship between two fundamental accounting constructs that reveals managerial reporting incentives and asset quality simultaneously. Our findings extend Hirshleifer et al. (2004)'s work on net operating assets by showing that the interaction between working capital and fixed asset accounting contains incremental information beyond level effects. Methodologically, we contribute to the anomalies literature by subjecting ADS to the comprehensive testing protocol of Novy-Marx and Velikov (2023), demonstrating its robustness across multiple dimensions including alternative portfolio constructions, transaction costs, and controls for related signals. The strategy's ability to generate significant alphas even after controlling for 158 existing anomalies in combination strategies underscores its unique information content. Our results have important implications for understanding market efficiency and the limits of arbitrage. The persistence of ADS-based return predictability, particularly among larger stocks where institutional participation is highest, suggests that even sophisticated investors struggle to fully process complex accounting interactions. This evidence supports theoretical models of gradual information diffusion and highlights the economic importance of accounting measurement choices in asset pricing, contributing to the broader debate on market efficiency and the role of financial reporting in capital markets.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT North America database, spanning several decades of annual observations. We focus on firms with sufficient data to construct our accrual depreciation sensitivity measure, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and accounting practices.signal construction relies on two key accounting variables from the statement of cash flows. Assets and liabilities other net change (AOLOCH) represents the net change in other assets and liabilities not separately disclosed in the cash flow statement, capturing residual working capital adjustments and other operating accruals that affect cash flows from operations. Depreciation, depletion, and amortization (DPACT) measures the total non-cash charges related to the allocation of tangible and intangible asset costs over their useful lives, representing a major component of the reconciliation between net income and operating cash flows.construct the Accrual Depreciation Sensitivity signal by dividing AOLOCH by DPACT for each firm-year observation. This ratio captures the relationship between other operating accruals and depreciation expenses, providing insight into how firms' accrual patterns scale with their non-cash asset charges. We calculate the signal using annual observations at each fiscal year-end, requiring non-missing and non-zero values for DPACT to avoid undefined ratios. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. The resulting measure reflects the sensitivity of residual accruals to depreciation levels, with higher values indicating larger other accrual changes relative to depreciation expenses.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ADS signal. Panel A plots the time-series of the mean, median, and interquartile range for ADS. On average, the cross-sectional mean (median) ADS is -0.09 (-0.00) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input ADS data. The signal's interquartile range spans -0.26 to 0.17. Panel B of Figure 1 plots the time-series of the coverage of the ADS signal for the CRSP universe. On average, the ADS signal is available for 6.67% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does ADS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ADS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ADS portfolio and sells the low ADS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ADS strategy earns an average return of 0.23% per month with a t-statistic of 2.41. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.23% to 0.28% per month and have t-statistics exceeding 2.37 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.11, with a t-statistic of -3.31 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 534 stocks and an average market capitalization of at least $2,604 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.41. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 3-32bps/month. The lowest return, (3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.36. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ADS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-two cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the ADS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ADS, as well as average returns and alphas for long/short trading ADS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ADS strategy achieves an average return of 23 bps/month with a t-statistic of 1.76. Among these large cap stocks, the alphas for the ADS strategy relative to the five most common factor models range from 16 to 33 bps/month with t-statistics between 1.27 and 2.58.

5 How does ADS perform relative to the zoo?

Figure 2 puts the performance of ADS in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

ratio for the ADS strategy falls in the distribution. The ADS strategy's gross (net) Sharpe ratio of 0.40 (0.35) is greater than 82% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the ADS strategy (red line).[2] Ignoring trading costs, a $1 invested in the ADS strategy would have yielded $1.57 which ranks the ADS strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a $1 invested in the ADS strategy would have yielded $1.27 which ranks the ADS strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ADS relative to those. Panel A shows that the ADS strategy gross alphas fall between the 50 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ADS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ADS ranks between the 77 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ADS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ADS with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ADS or at least to weaken the power ADS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ADS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ADS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ADS} ADS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ADS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ADS. Stocks are finally grouped into five ADS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ADS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ADS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ADS signal in these Fama-MacBeth regressions exceed 1.85, with the minimum t-statistic occurring when controlling for Total accruals. Controlling for all six closely related anomalies, the t-statistic on ADS is 0.60.

Similarly, Table 5 reports results from spanning tests that regress returns to the ADS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ADS strategy earns alphas that range from 19-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.01, which is achieved when controlling for Total accruals. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ADS trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.51.

7 Does ADS add relative to the whole zoo?

Finally, we can ask how much adding ADS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ADS signal.[4] We consider one different methods for combining signals.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ADS is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $36.86, while $1 investment in the combination strategy that includes ADS grows to $34.52.

8 Conclusion

This study provides compelling evidence that Accrual Depreciation Sensitivity (ADS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, employing the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that ADS-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.40 gross (0.35 net of transaction costs), indicating strong economic viability. Most notably, the strategy delivers statistically significant alphas of 25 bps/month relative to the Fama-French five-factor model plus momentum, with this performance remaining robust at 23 bps/month even when controlling for six closely related strategies including total accruals and abnormal accruals. These findings suggest that ADS captures unique information about firm fundamentals not fully reflected in existing pricing factors or accrual-based anomalies. The practical implications are substantial for both academic researchers and investment practitioners. For academics, ADS enriches our understanding of how accounting information, particularly the interaction between accruals and depreciation policies, influences asset prices. For practitioners, the strategy's resilience to transaction costs (maintaining a Sharpe ratio of 0.35 after costs) and its alpha generation relative to multiple factor models suggest potential implementation value in quantitative investment strategies. However, several limitations warrant consideration. First, while our results are robust to standard risk adjustments, the possibility of unidentified risk factors cannot be entirely ruled out. Second, the implementation of ADS strategies may face practical challenges including data availability, particularly for smaller firms, and potential capacity constraints in real-world trading environments. Third, as markets evolve and become more efficient, the persistence of this anomaly remains an empirical question. Future research should explore several promising avenues. Investigating the economic mechanisms underlying the ADS effect could provide deeper insights into why markets appear to misprice this information. Cross-country studies would help establish whether ADS represents a global phenomenon or is specific to certain market structures. Additionally, examining the interaction between ADS and corporate events such as earnings announcements or management changes could reveal when the signal is most informative. Finally, exploring machine learning techniques to potentially enhance the ADS signal or combine it optimally with other predictors represents a natural extension of this work. In conclusion, our findings establish ADS as a valuable addition to the factor zoo, one that survives stringent empirical scrutiny and offers both theoretical insights and practical applications for understanding and exploiting cross-sectional return predictability.

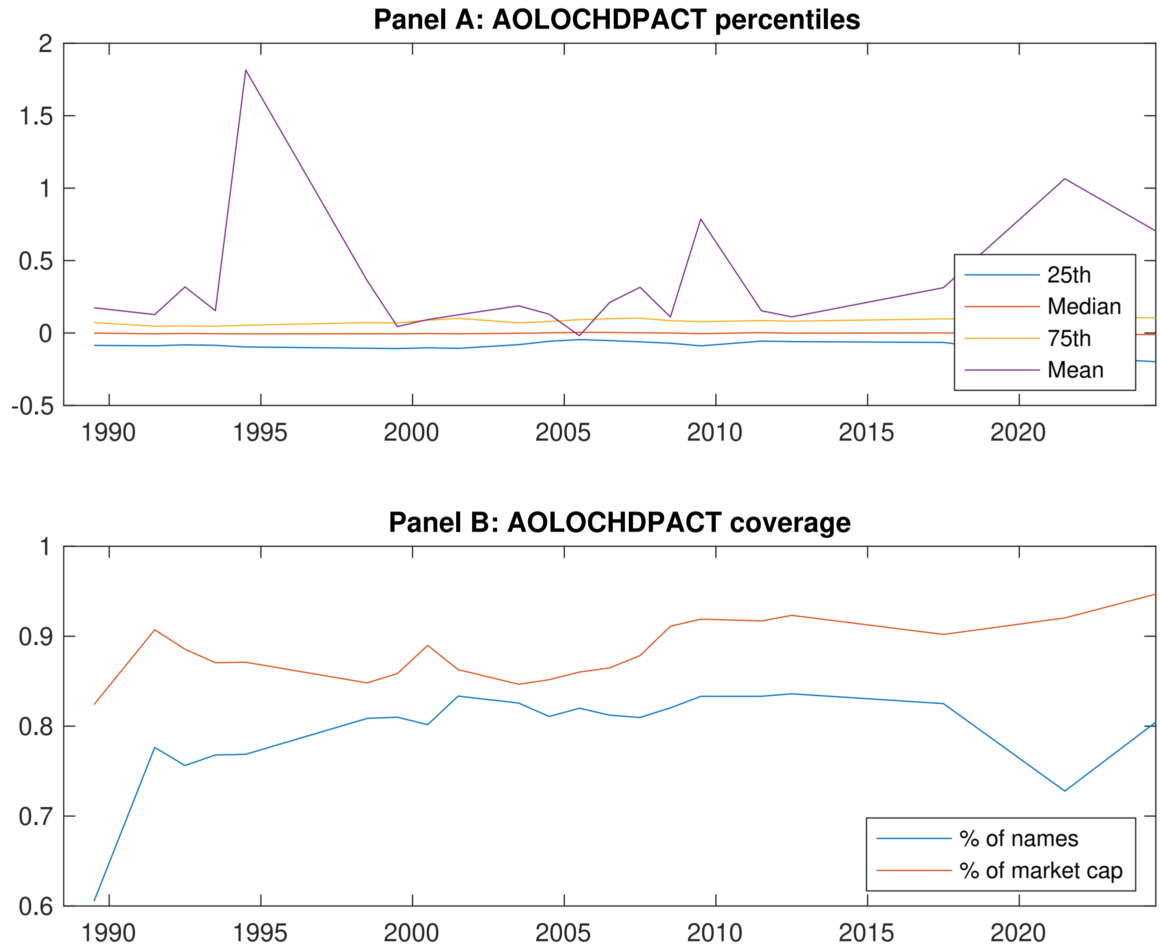


Figure 1: Times series of ADS percentiles and coverage.
This figure plots descriptive statistics for ADS. Panel A shows cross-sectional percentiles of ADS over the sample. Panel B plots the monthly coverage of ADS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ADS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on ADS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.68	0.65	0.73	0.70	0.91	0.23
	[2.75]	[3.18]	[3.83]	[3.58]	[3.79]	[2.41]
α_{CAPM}	-0.16	-0.04	0.11	0.04	0.10	0.26
	[-2.28]	[-0.59]	[1.45]	[0.66]	[1.39]	[2.65]
α_{FF3}	-0.13	-0.04	0.06	0.03	0.14	0.27
	[-1.95]	[-0.68]	[0.97]	[0.58]	[2.31]	[2.82]
α_{FF4}	-0.08	-0.02	0.06	0.02	0.14	0.23
	[-1.26]	[-0.33]	[0.92]	[0.29]	[2.36]	[2.37]
α_{FF5}	-0.04	-0.15	-0.09	-0.08	0.24	0.28
	[-0.64]	[-2.37]	[-1.36]	[-1.50]	[4.13]	[2.94]
α_{FF6}	-0.01	-0.12	-0.08	-0.09	0.24	0.25
	[-0.14]	[-1.93]	[-1.23]	[-1.56]	[4.00]	[2.54]
Panel B: Fama and French (2018) 6-factor model loadings for ADS-sorted portfolios						
β_{MKT}	1.07	0.96	0.92	0.95	1.04	-0.02
	[65.14]	[61.17]	[57.18]	[69.21]	[70.20]	[-0.97]
β_{SMB}	0.09	0.02	-0.01	-0.02	-0.03	-0.11
	[3.82]	[0.95]	[-0.28]	[-1.16]	[-1.21]	[-3.31]
β_{HML}	-0.09	-0.13	0.04	-0.12	-0.08	0.00
	[-3.11]	[-4.69]	[1.47]	[-5.08]	[-3.27]	[0.10]
β_{RMW}	-0.14	0.11	0.24	0.13	-0.10	0.04
	[-4.70]	[4.01]	[8.36]	[5.38]	[-3.82]	[0.84]
β_{CMA}	-0.07	0.27	0.20	0.24	-0.26	-0.19
	[-1.70]	[6.66]	[4.99]	[6.73]	[-6.89]	[-3.07]
β_{UMD}	-0.05	-0.04	-0.01	0.01	0.01	0.06
	[-3.61]	[-3.11]	[-0.80]	[0.56]	[0.56]	[2.77]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1001	620	534	606	970	
me ($\$10^6$)	2604	2650	2820	3153	4024	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ADS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23	0.26	0.27	0.23	0.28	0.25
			[2.41]	[2.65]	[2.82]	[2.37]	[2.94]	[2.54]
Quintile	NYSE	EW	0.24	0.27	0.27	0.21	0.27	0.21
			[3.87]	[4.32]	[4.43]	[3.49]	[4.16]	[3.41]
Quintile	Name	VW	0.35	0.38	0.39	0.35	0.45	0.41
			[3.23]	[3.38]	[3.70]	[3.23]	[4.19]	[3.76]
Quintile	Cap	VW	0.27	0.24	0.27	0.23	0.34	0.31
			[2.58]	[2.34]	[2.63]	[2.27]	[3.32]	[2.94]
Decile	NYSE	VW	0.34	0.35	0.38	0.36	0.45	0.42
			[2.72]	[2.83]	[3.21]	[2.98]	[3.81]	[3.54]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.20	0.24	0.25	0.22	0.26	0.23
			[2.12]	[2.46]	[2.62]	[2.37]	[2.69]	[2.46]
Quintile	NYSE	EW	0.03	0.06	0.05			
			[0.36]	[0.77]	[0.73]			
Quintile	Name	VW	0.32	0.37	0.39	0.36	0.43	0.41
			[2.94]	[3.38]	[3.70]	[3.45]	[4.11]	[3.87]
Quintile	Cap	VW	0.24	0.23	0.25	0.23	0.31	0.29
			[2.32]	[2.18]	[2.44]	[2.25]	[3.03]	[2.82]
Decile	NYSE	VW	0.30	0.32	0.34	0.33	0.40	0.38
			[2.43]	[2.56]	[2.88]	[2.77]	[3.35]	[3.22]

Table 3: Conditional sort on size and ADS

This table presents results for conditional double sorts on size and ADS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ADS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ADS and short stocks with low ADS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results

	ADS Quintiles					ADS Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.35	0.80	0.96	1.00	0.57	0.22	0.24	0.23	0.17	0.26	0.21
	[0.95]	[2.46]	[2.92]	[3.14]	[1.59]	[2.00]	[2.15]	[2.06]	[1.55]	[2.22]	[1.80]
(2)	0.61	0.77	0.89	0.84	0.62	0.00	0.09	0.09	0.05	0.06	0.03
	[1.81]	[2.67]	[3.20]	[2.92]	[1.96]	[0.04]	[0.80]	[0.78]	[0.46]	[0.51]	[0.26]
(3)	0.67	0.77	0.79	0.81	0.83	0.16	0.16	0.17	0.17	0.25	0.24
	[2.19]	[2.91]	[3.15]	[3.19]	[2.73]	[1.42]	[1.37]	[1.51]	[1.47]	[2.17]	[2.06]
(4)	0.77	0.75	0.80	0.73	0.98	0.21	0.20	0.26	0.21	0.38	0.33
	[2.83]	[3.14]	[3.43]	[3.13]	[3.46]	[1.88]	[1.80]	[2.56]	[2.04]	[3.71]	[3.21]
(5)	0.71	0.57	0.75	0.72	0.94	0.23	0.16	0.20	0.17	0.33	0.29
	[3.11]	[2.87]	[4.01]	[3.63]	[3.70]	[1.76]	[1.27]	[1.58]	[1.31]	[2.58]	[2.26]

Panel B: Portfolio average number of firms and market capitalization

	ADS Quintiles Average n					ADS Quintiles Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	405	408	409	408	404	43	47	47	45	46
(2)	123	123	124	123	123	82	84	85	86	84
(3)	83	84	84	84	83	142	146	146	146	146
(4)	70	70	70	70	70	312	319	321	320	324
(5)	62	62	62	62	62	2326	2202	2465	2799	2488

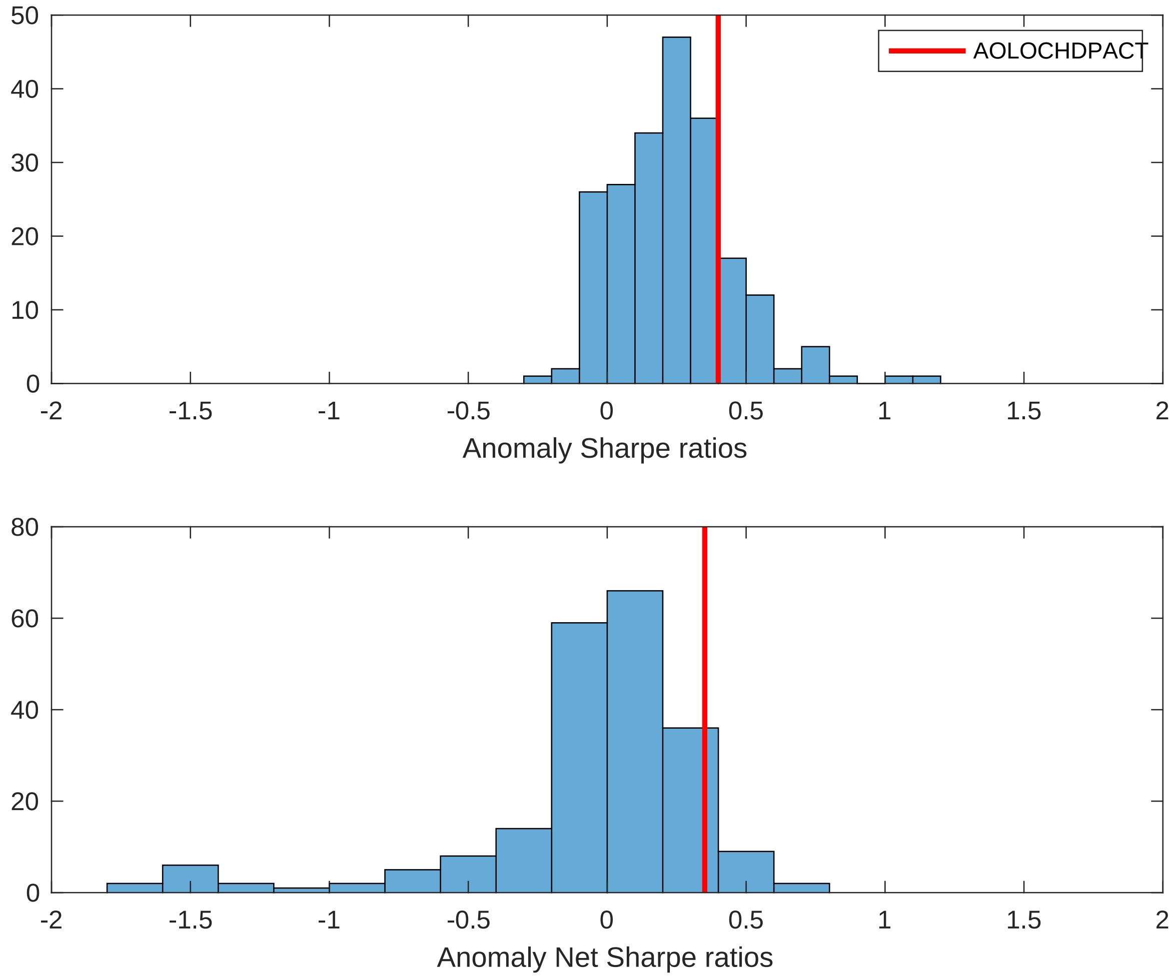


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ADS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

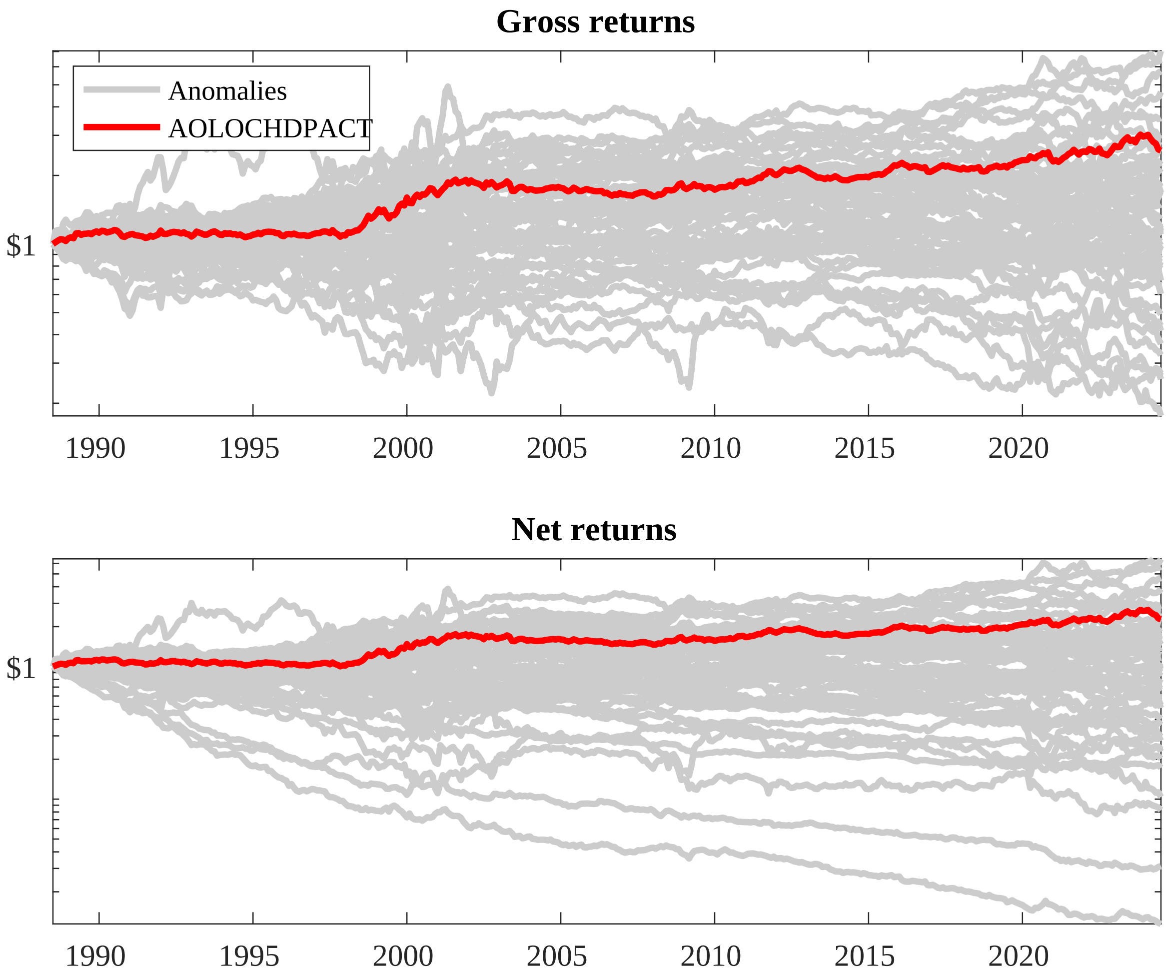
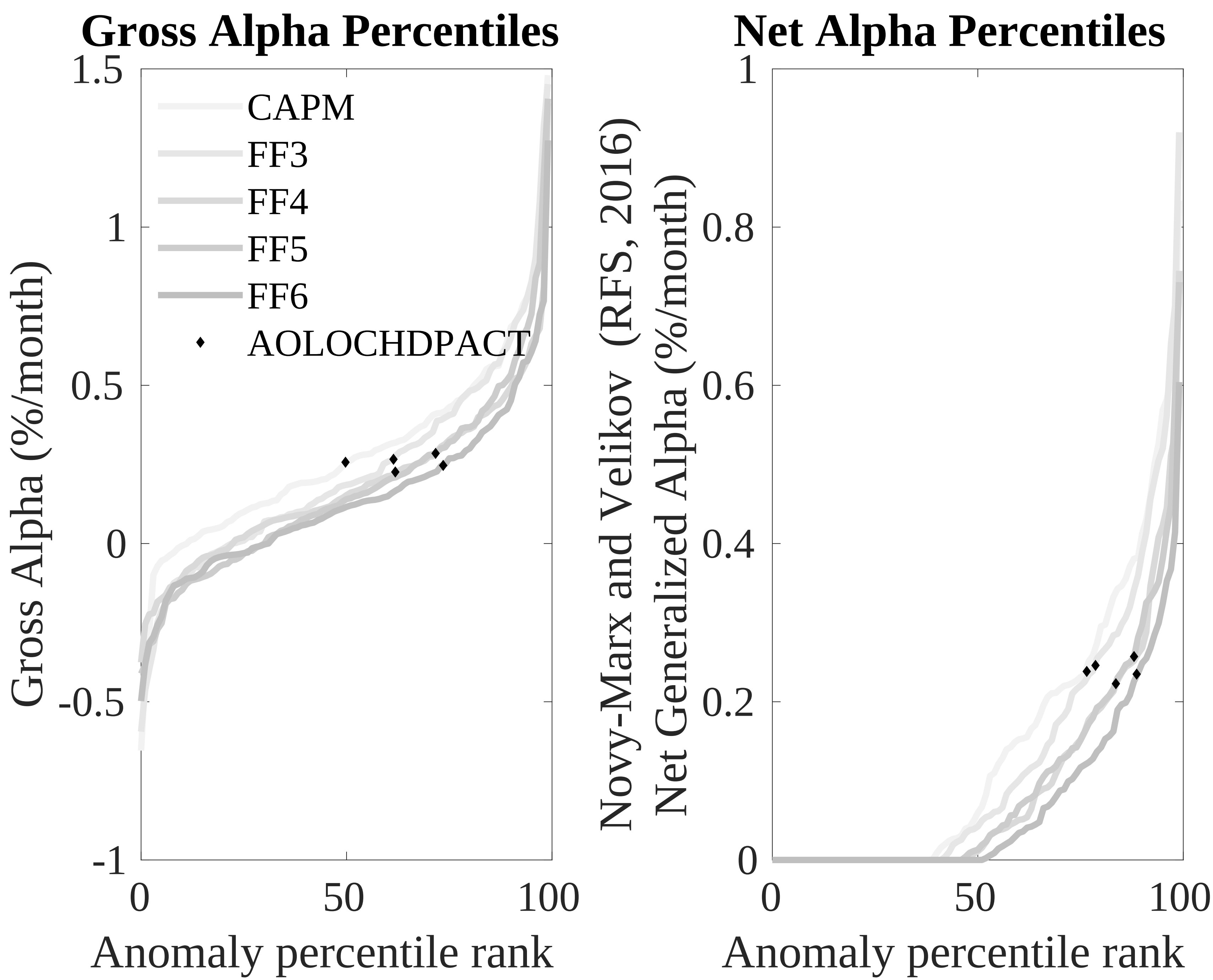
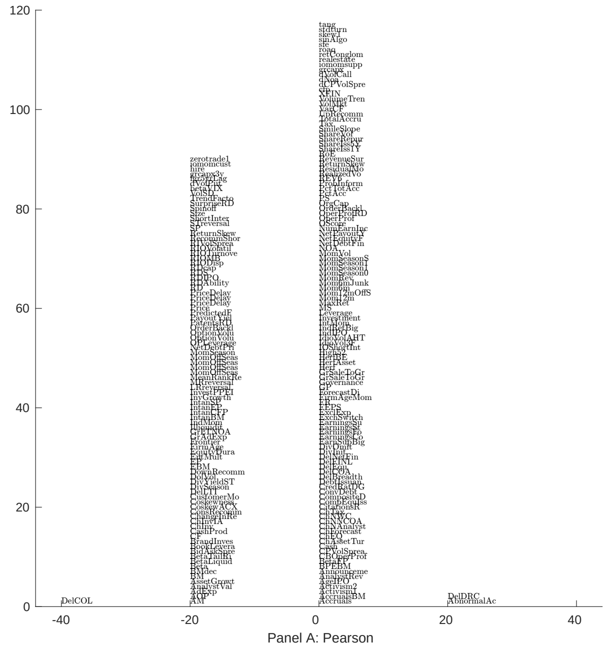


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ADS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
AOLOCHDPACT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



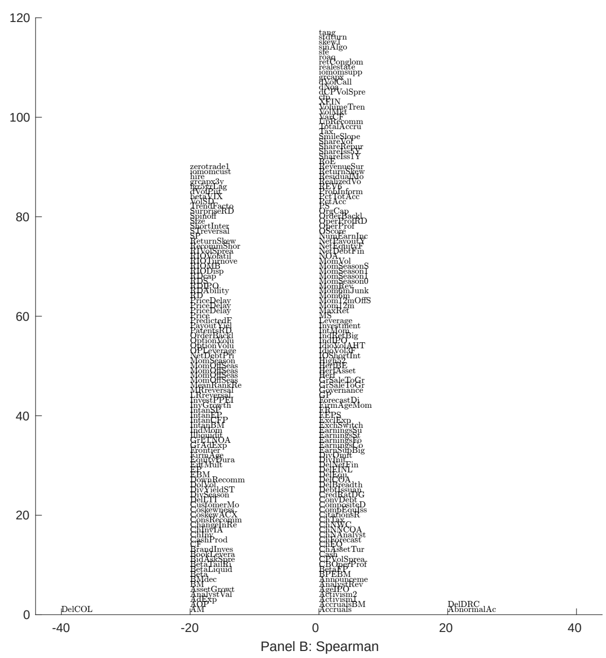


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with ADS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

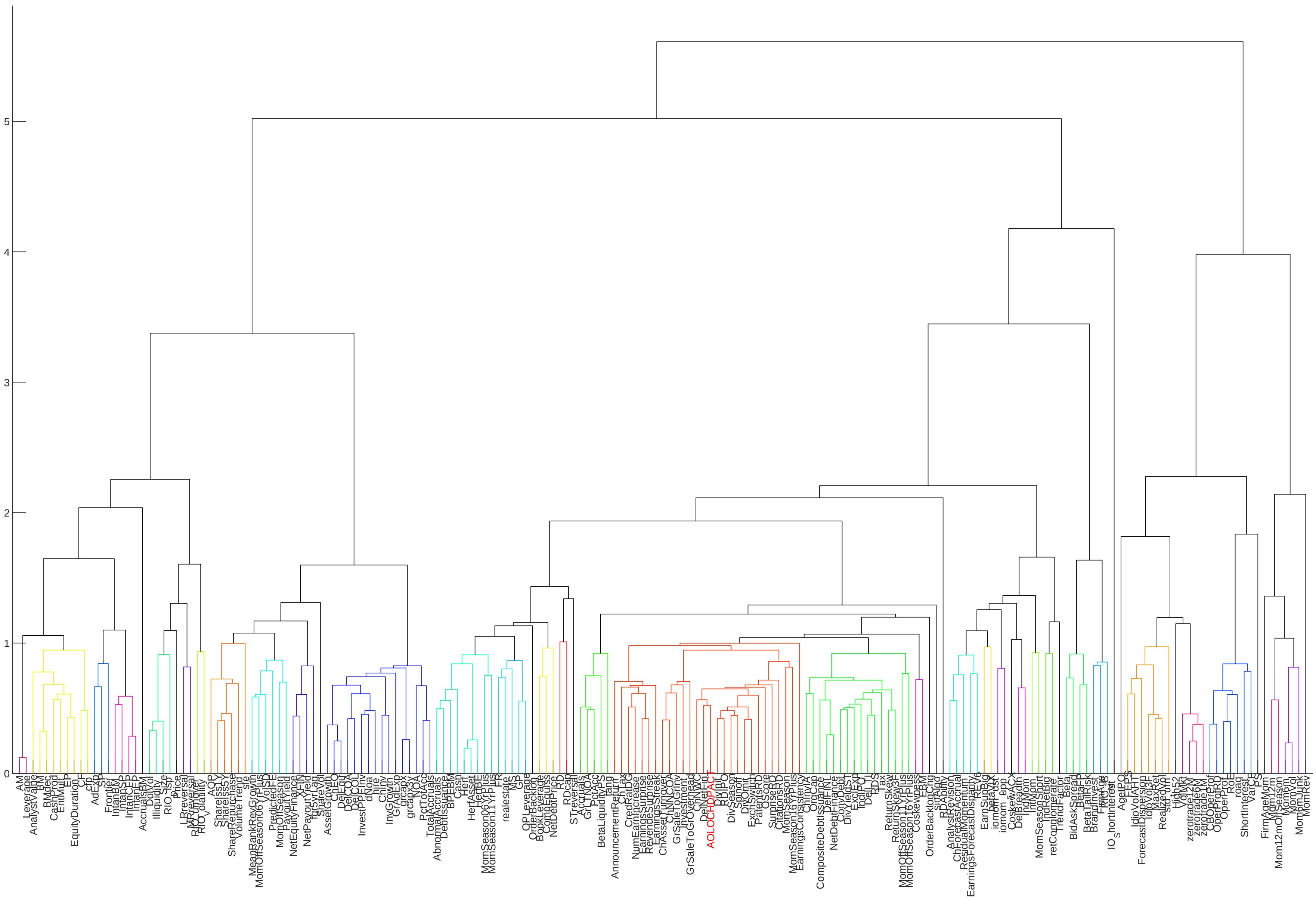


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

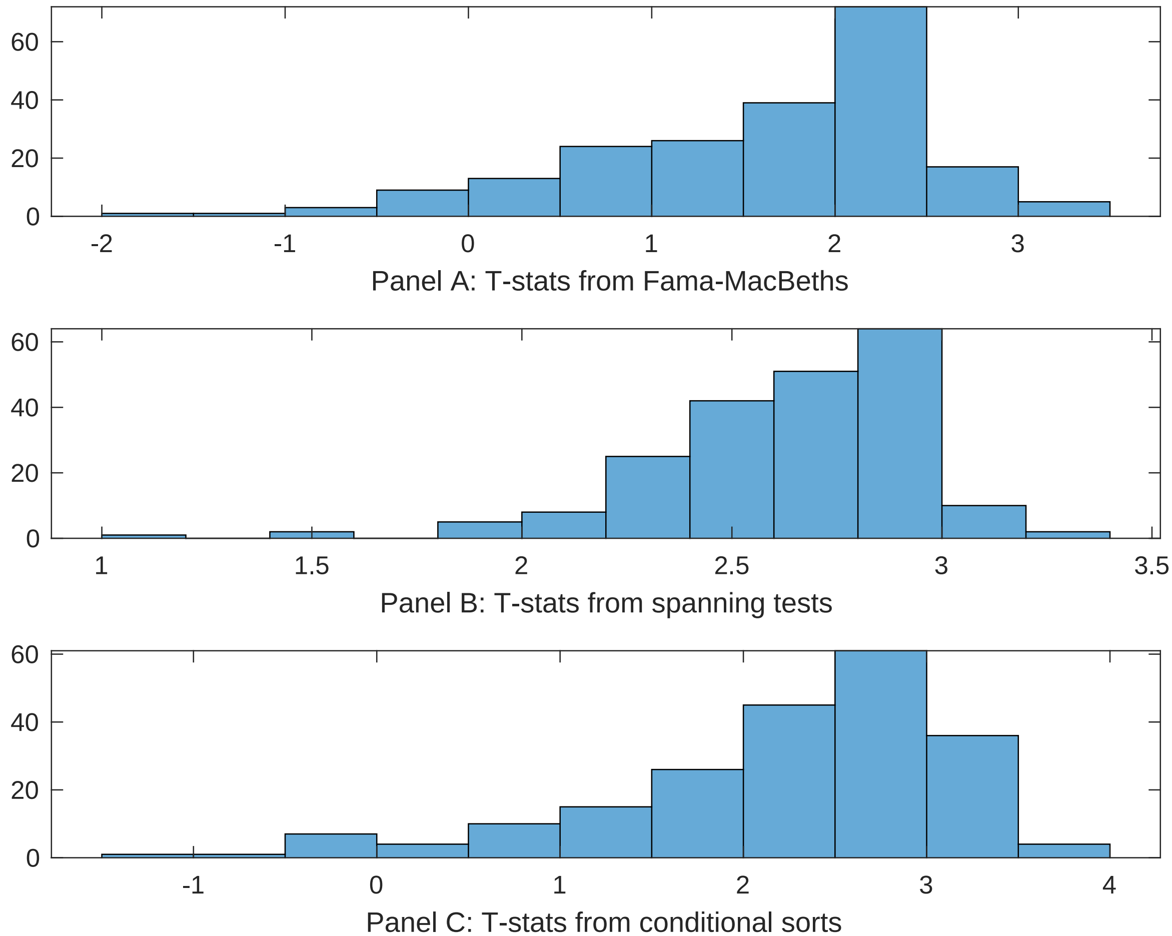


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ADS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ADS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ADS} ADS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ADS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ADS. Stocks are finally grouped into five ADS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ADS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ADS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ADS} ADS_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11	0.99	0.11	0.11	0.12	0.11	0.12
	[3.89]	[3.37]	[3.87]	[3.46]	[3.89]	[3.09]	[3.55]
ADS	0.29	0.60	0.37	0.33	0.57	0.37	0.30
	[1.85]	[2.14]	[2.32]	[2.25]	[1.95]	[2.62]	[0.60]
Anomaly 1	0.50						0.79
	[1.85]						[2.76]
Anomaly 2		0.16					-0.74
		[0.24]					[-0.79]
Anomaly 3			0.51				-0.12
			[0.18]				[-0.27]
Anomaly 4				0.42			-0.21
				[0.26]			[-0.20]
Anomaly 5					0.47		0.37
					[4.84]		[4.33]
Anomaly 6						0.62	-0.29
						[1.17]	[-0.65]
# months	432	427	432	432	427	432	415
$\bar{R}^2$(%)	0	1	0	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the ADS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ADS} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.27	0.30	0.19	0.26	0.29	0.26	0.23
	[2.89]	[3.12]	[2.01]	[2.73]	[3.07]	[2.66]	[2.51]
Anomaly 1	-19.87						-16.13
	[-4.53]						[-3.70]
Anomaly 2		-5.26					-4.55
		[-1.81]					[-1.60]
Anomaly 3			21.17				19.96
			[4.79]				[4.53]
Anomaly 4				-3.41			-11.20
				[-0.64]			[-1.68]
Anomaly 5					-0.63		2.99
					[-0.22]		[0.96]
Anomaly 6						3.28	5.38
						[0.88]	[1.12]
mkt	-2.43	-2.25	-5.10	-3.06	-2.55	-3.31	-6.04
	[-1.03]	[-0.95]	[-2.13]	[-1.25]	[-1.07]	[-1.34]	[-2.31]
smb	-14.32	-10.71	-14.20	-9.01	-12.70	-15.83	-7.30
	[-4.21]	[-2.95]	[-4.19]	[-1.31]	[-3.45]	[-3.26]	[-1.10]
hml	-2.50	1.85	2.17	-0.23	0.03	-0.94	2.02
	[-0.60]	[0.43]	[0.52]	[-0.05]	[0.01]	[-0.22]	[0.47]
rmw	-0.91	5.70	6.09	2.50	4.27	5.39	5.64
	[-0.21]	[1.31]	[1.42]	[0.55]	[0.99]	[1.08]	[1.13]
cma	-4.58	-17.66	-13.55	-18.30	-19.77	-18.81	-3.64
	[-0.68]	[-2.86]	[-2.23]	[-2.97]	[-3.24]	[-3.04]	[-0.54]
umd	4.02	3.72	4.78	4.92	5.31	7.87	4.62
	[1.89]	[1.57]	[2.28]	[1.98]	[1.86]	[2.42]	[1.36]
# months	432	428	432	432	428	432	428
$\bar{R}^2(\%)$	14	11	14	10	11	10	18

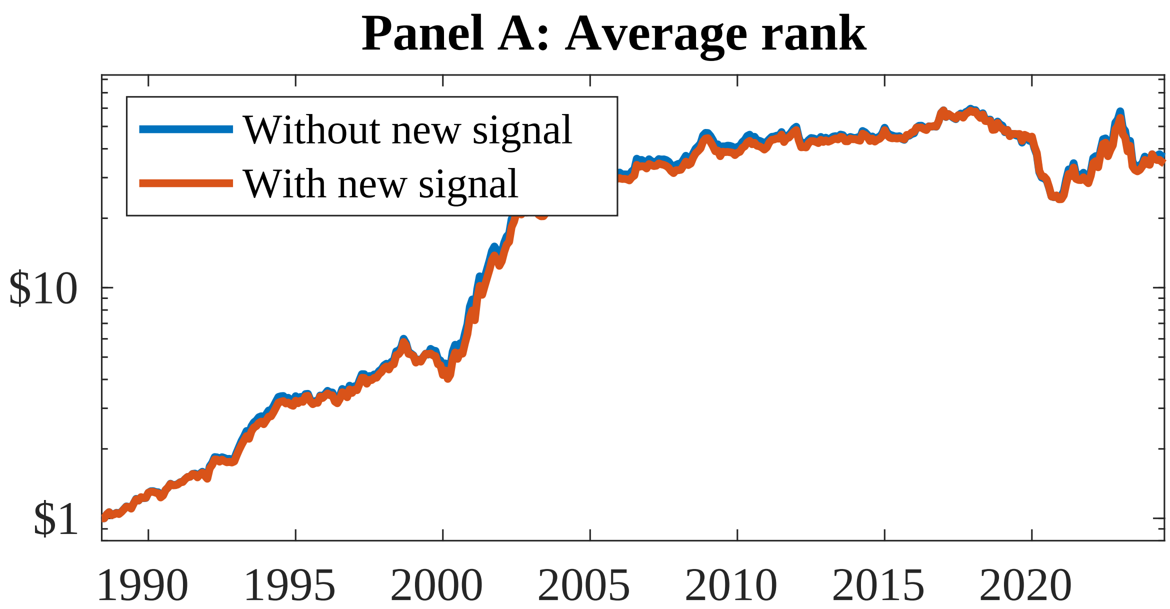


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ADS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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